Intrinsic-Noise Consolidation: A Doob-Barrier-Conditioned Diffusion Turns Analog Device Noise into a Continual-Learning Resource

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Abstract

On analog neuromorphic hardware, intrinsic device noise is normally an accuracy tax. We ask whether it can instead be made to *consolidate* memories. We cast per-synapse consolidation as a *Doob h-transform*: condition each weight's stochastic dynamics on the event of never crossing a memory-critical barrier around its consolidated value. The conditioned diffusion acquires an extra drift $\sigma^2\,\partial_w \log h$ — a restoring force toward the memory that is *amplified by the noise variance itself*, and diverges at the barrier. We are explicit about what is and is not new. The anchored-consolidation drift $-s(w-\mu)$ that our rule also contains is *not* ours: it is the small-noise limit of Ornstein–Uhlenbeck Adaptation [Garcia Fernandez et al., 2024], the variance-scaled anchor of MESU [Bonnet et al., 2025], and the Fisher penalty of EWC [Kirkpatrick et al., 2017], and we surrender it as a re-derivation. Our claim is the conjunction of (a) the Doob barrier-conditioning as a synaptic rule — to our knowledge unclaimed; every h-transform use we found is generative modeling or Schrödinger bridges, *none* synaptic — and (b) a falsifiable, load-bearing prediction: increasing the intrinsic noise *non-monotonically* improves sequential-task retention, an inverted-U these anchored-drift methods cannot produce. We pre-registered this as a go/no-go gate and it passes: on single-head Split-MNIST (8 seeds) the barrier-conditioned rule lifts retention by 10.9 percentage points at an interior optimum $\sigma^* = 0.02$ (paired Wilcoxon $p = 0.004$ vs. zero noise and vs. high noise), while the matched OU, EWC and MESU anchors are monotone-decreasing in noise. Ablating the conditioning removes the effect; the optimum tracks the barrier; the inverted-U survives a device-faithful BrainScaleS-2 noise model (colored, multiplicative, fixed-pattern, 6-bit) and reproduces on a second task stream. At its optimum the rule is the strongest *rehearsal-free* consolidation method we test — matching MESU and significantly beating OU, EWC and Benna–Fusi; plain replay, which stores data, scores higher but exhibits none of the mechanism. We do *not* claim measured silicon: our BrainScaleS-2 result is an emulation, and an on-chip reproduction with measured energy is the pre-registered remaining step. Within these limits the mechanism reframes analog noise from a tax into a consolidation dividend that a von-Neumann accelerator must spend energy to fake.

1 Introduction

Catastrophic forgetting — the overwriting of old memories when a network learns something new — is usually fought with regularizers that anchor important weights to their consolidated values [Kirkpatrick et al., 2017, Benna and Fusi, 2016]. On digital hardware, stochasticity is something you add deliberately and pay for. On *analog* neuromorphic substrates such as BrainScaleS-2 (BSS-2) the situation is inverted: the hardware is *intrinsically* noisy — thermal fluctuations, crosstalk, analog-storage drift [Weis et al., 2020, Pehle et al., 2022] — and this noise is normally treated as an accuracy tax to be calibrated away. This paper asks a contrarian question: can that same intrinsic noise be *steered* so that it consolidates memories rather than degrading them?

One idea, stated as an identity plus a conjecture. Consider a single synaptic weight w that, after learning a task, should stay near a consolidated value μ to preserve that task's function. Model its dynamics as a diffusion. If we simply pull it back — $dw = -s(w-\mu)\,dt + \sigma\,dW$ — we recover a mean-reverting Ornstein–Uhlenbeck (OU) process whose stationary spread $\sigma^2/2s$ *grows* with the noise: more noise is strictly worse. This anchored drift is exactly the rule already published as OU Adaptation [Garcia Fernandez et al., 2024], the variance-scaled Bayesian anchor of MESU [Bonnet et al., 2025], and the small-step limit of EWC [Kirkpatrick et al., 2017]; we claim no novelty for it and say so throughout. Our move is different: we *condition* the diffusion on the event that w never crosses a memory-critical barrier at $\mu \pm b$. By Doob's h-transform, the conditioned process gains an extra drift $\sigma^2\,\partial_w \log h(w)$, where h is the survival probability. This term (i) points *into* the safe region, (ii) *diverges* at the barrier, and — crucially — (iii) scales with σ^2: *the more intrinsic noise, the stronger the noise-powered restoring force.* The competition between this σ^2 steering (which, at moderate noise, best resists interference) and the raw σ diffusion (which, at high noise, overwhelms the steering) predicts a non-monotone, inverted-U relationship between noise and retention with an optimum at $\sigma > 0$.

The claim split (and why it matters). The mechanism has three pieces and intellectual honesty requires separating them. We **surrender** the drift: $-s(w-\mu)$ is a known limit of OUA/MESU/EWC, proven so, not merely "similar." We **keep**, as the entire contribution, the conjunction of two things: (a) casting per-synapse consolidation as a Doob / barrier-conditioned diffusion — an unclaimed framing (every h-transform use we located is generative modeling, Schrödinger bridges, or transition-path sampling [Du et al., 2025, Nguyen et al., 2025, Deng et al., 2024], *zero* synaptic) — and (b) the falsifiable hardware curve: intrinsic noise non-monotonically improving sequential-task retention. Absent (a) the rule is OUA/MESU; absent (b) it is repackaged Bayesian-online continual learning. The paper lives or dies on (a) *and* (b). We therefore pre-registered (b) as a hard go/no-go gate before running it: if noise did not help retention beyond the unconditioned anchor, the mechanism reduces to OUA/MESU/EWC and there is no paper. It helps.

Contributions.

1. **A synaptic Doob h-transform (novel framing).** We derive the per-synapse barrier-conditioned rule from the ground-state h-transform of the iso-loss interval, with the memory-critical barrier read off the Fisher metric (§3). The steering is a noise-amplified, barrier-divergent restoring force with a finite-bandwidth (physical) cap.

2. **A pre-registered falsifier that passes (load-bearing).** On Split-MNIST (8 seeds) the rule produces a retention inverted-U, lift 10.9 pts at $\sigma^* = 0.02$ ($p = 0.004$), while matched OU, EWC and MESU anchors are monotone in noise (§4.1, Fig. 1).

3. **Mechanism isolation.** Ablating the conditioning ($\kappa : 1 \to 0$) flattens the curve; the optimum tracks the barrier scale (§4.2). The effect is the conditioning, not generic noise.

4. **Device-faithful robustness and a second modality.** The inverted-U survives a BSS-2 intrinsic-noise emulation (colored + multiplicative + fixed-pattern + 6-bit) and reproduces on continual Yin-Yang (§4.3,4.5). We are explicit that this is emulation, not silicon.

5. **Baselines and an energy argument.** At its optimum the rule is the strongest *rehearsal-free* consolidation method we test (ties MESU, beats OU/EWC/Benna–Fusi); replay, which stores data, does better but lacks the mechanism. Because the diffusion is the device's own

noise, an analog substrate pays no energy to generate it, whereas a digital accelerator does (§4.4).

The distinguishing statement. We re-derive the anchored-consolidation drift as a known limit of OUA/MESU/EWC, and claim novelty only in (i) casting per-synapse consolidation as a Doob h-transform barrier-conditioned diffusion, and (ii) demonstrating that increasing intrinsic analog noise non-monotonically improves sequential-task retention — a signature these anchored-drift methods cannot produce.

2 Related work

The anchored drift is not ours. **OUA** [Garcia Fernandez et al., 2024] frames learning as the mean-reverting OU diffusion $d\theta = \lambda(\mu - \theta)dt + \Sigma\, dW$ — exactly our anchored drift — but as a reward-modulated learning rule; it names catastrophic forgetting only as future work and uses no barrier or first-passage conditioning. **MESU** [Bonnet et al., 2025] is Bayesian continual learning whose update (their Eq. 11) is a variance-scaled pull toward a prior mean; it treats device read-noise as a *sampling* resource for Monte-Carlo Bayesian updates, never as a retention optimum, and uses no Doob transform. **EWC** [Kirkpatrick et al., 2017] is the static Fisher-weighted quadratic anchor, the deterministic ancestor. All three are the drift's origin; none contains (a) or (b).

Noise as a resource, but not this one. Kolesnikov and Semenova [2025] show internal hardware noise can *help* — but with resilience to test-time noise in *single-task* feedforward/echo-state nets, not sequential-task retention, and with no inverted-U over retention. Shaham et al. [2022] consolidate lifelong memory with stochastic *rehearsal* (a replay mechanism), not intrinsic per-synapse device noise, and with no hardware or barrier. NADO [Manneschi et al., 2025] trains *through* device stochasticity with neural-SDE digital twins for single-task temporal tasks. Probabilistic Metaplasticity [Zohora et al., 2024] consolidates on memristors by modulating update *probability* (a stochastic gate) while treating device noise as an obstacle. None claims that raising intrinsic noise non-monotonically improves cross-task retention. **Benna–Fusi** complex synapses [Benna and Fusi, 2016] consolidate without a barrier via a deterministic multi-timescale cascade — an incumbent we compare against.

Doob's h-transform lives in generative modeling. Conditioning a diffusion on a rare event via the h-transform underlies transition-path sampling [Du et al., 2025], diffusion-based image editing [Nguyen et al., 2025], constrained/reflected Schrödinger bridges [Deng et al., 2024], and diffusion-bridge simulation [Heng et al., 2021]. We found *no* use of Doob's h-transform as a synaptic, plasticity, or continual-learning rule. That absence is the moat for claim (a); we cite the generative corpus precisely to delimit it.

3 Method

Setup. A network with weights θ learns a sequence of tasks. After task t we store an anchor $\mu = \theta_t$ and an importance s = online-EWC running sum of the model-sampled (true) diagonal Fisher. During a later task each weight follows, under Euler–Maruyama,

$$dw_i = \Big[\underbrace{-\,\partial_{w_i}\mathcal{L}_{\text{task}}}_{\text{learning}} \quad \underbrace{-\,s_i\,(w_i - \mu_i)}_{\text{anchored drift (surrendered)}} \quad + \quad \underbrace{\sigma_i^2\,\partial_{w_i}\log h_i(w)}_{\text{Doob steering (ours)}}\Big]\,dt \;+\; \underbrace{\sigma_i\,dW_i}_{\text{intrinsic noise}}\;. \qquad (1)$$

Every method we compare shares the first term and receives the *identical* injected noise at matched σ; methods differ only in their drift. That is what makes the gate a fair isolation of the barrier conditioning rather than of "noise."

The barrier-conditioned (Doob) steering. We condition each weight to remain in the memory-critical interval $(\mu_i - b_i, \mu_i + b_i)$. The ground-state (quasi-stationary) h-transform of Brownian motion killed at the interval ends has $h_i(w) = \cos\big(\frac{\pi(w-\mu_i)}{2b_i}\big)$, so

$$\partial_w \log h_i = -\frac{\pi}{2b_i}\,\tan\Big(\frac{\pi(w-\mu_i)}{2b_i}\Big), \tag{2}$$

a restoring force toward the anchor that diverges at $\mu_i \pm b_i$ and, in Eq. (1), is multiplied by σ_i^2: the intrinsic noise powers the confinement. The barrier half-width is the softened iso-loss radius $b_i = b_0/\sqrt{1 + s_i/\mathrm{median}(s)}$, so important (high-Fisher) synapses get a tight barrier and unimportant ones a loose one at $\sim b_0$; the same global noise is thereby *steered per synapse* by the memory geometry. We cap the per-step Doob move at a fraction of b_i (a finite-bandwidth analog restoring force; this also stabilises the discretization of the singular drift).

Matched controls and incumbents. *OU* is Eq. (1) with the Doob term deleted (the ablation; steering strength κ=0). *EWC* is a stronger static anchor; *MESU* a variance-scaled anchor (Eq. 11 of Bonnet et al., 2025, in spirit); *naive* is plain SGD. All receive the same injected noise. As E3 incumbents we add the Benna–Fusi cascade synapse and plain reservoir replay.

Testbeds. Primary: *Split-MNIST*, domain-incremental — five binary tasks (0v1, …, 8v9) on one shared 2-way head (later tasks overwrite the readout unless consolidation intervenes), an MLP 784–100–100–2. Second modality: *continual Yin-Yang* [Kriener et al., 2022], the BrainScaleS group's own procedural benchmark, as five rotations of the pattern on a shared 3-way head. Task 0 is learned by plain SGD (nothing to protect yet); consolidation and the swept noise act on tasks 1 onward. Retention is the mean final accuracy on the past tasks; we also report plasticity (accuracy just after learning each task) and forgetting.

BrainScaleS-2 noise model and the honesty boundary. For E2 we replace the white diffusion in Eq. (1) with a device-faithful BSS-2 noise model built from the published characterization [Weis et al., 2020, Pehle et al., 2022]: temporally *colored* (AR(1)) trial-to-trial variability, a *multiplicative* (signal-dependent) component, a static *fixed-pattern* per-synapse offset, and 6-bit weight quantization. This is an *emulation* on a GPU. We measure no silicon and no joules. Our energy statements come from a transparent operation-count model (per-operation constants from Horowitz, 2014); the load-bearing point is a ratio, robust to the constants: on a digital accelerator the diffusion noise must be *generated* (an RNG draw and a scaled add per weight per step) and the barrier drift computed, whereas on analog silicon the same stochastic term is the device's own physics — free. The on-chip reproduction with measured energy, via the port outlined in our code (hxtorch/pynn onto the analog array), is the pre-registered remaining step; the software stack is absent in our environment, so we neither run it nor claim it.

4 Experiments

We pre-registered the operating point (one coarse calibration, before the gated seeds), the noise grid, the seed counts, the inverted-U test, and the kill conditions K1–K3 in `PLAN.md`, committed

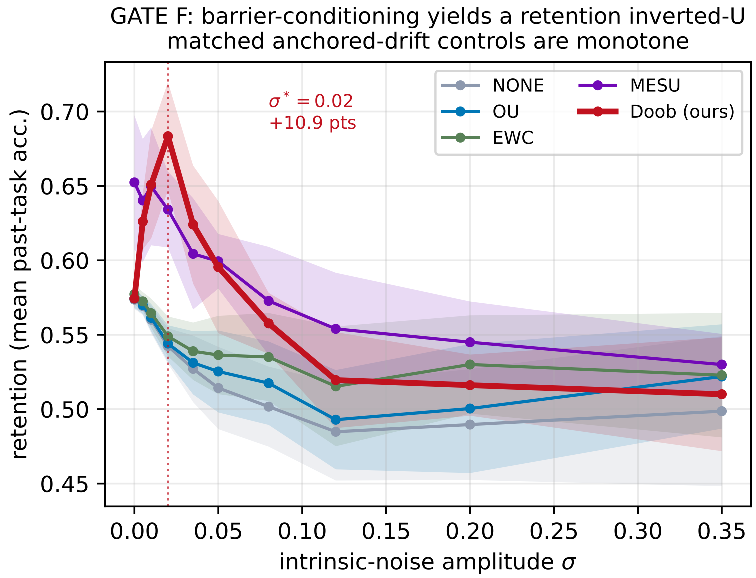

Figure 1: **GATE F (Split-MNIST, 8 seeds, mean±sd).** The barrier-conditioned rule (Doob, red) rises from 57.4% at zero noise to 68.3% at an interior optimum $\sigma^* = 0.02$ (a 10.9-point lift; paired Wilcoxon $p = 0.004$ vs. zero, $p = 0.004$ vs. the high-noise end), then falls. The matched OU, EWC, MESU and naive anchors are monotone-decreasing in noise. Noise is a *resource* only when steered by the barrier.

before any results.

4.1 E0 — GATE F: the pre-registered falsifier

We sweep the intrinsic-noise amplitude σ over 10 levels and measure retention for the barrier-conditioned rule and the matched anchored-drift controls, 8 seeds. GATE F passes iff the rule's retention–σ curve is an inverted-U (interior optimum $\sigma^* > 0$; peak beats both the zero-noise and high-noise ends by one-sided paired Wilcoxon; lift exceeds the seed spread) and no control is.

Result (Fig. 1). The rule is an inverted-U: retention rises from 57.4% (zero noise) to 68.3% at $\sigma^* = 0.02$, a lift of 10.9 points, then falls; both paired Wilcoxon tests give $p = 0.004$ (all 8 seeds agree). Every matched control is monotone-decreasing in σ (fraction of decreasing steps ≥ 0.78), peaking at zero noise. **GATE F passes: GO.** Because $\sigma^* > 0$ and the controls share the drift and the injected noise, the retention gain is attributable to the σ^2 barrier steering, not to noise per se — the mechanism does not reduce to OUA/MESU/EWC (kill condition K1 does not fire).

4.2 E1 — mechanism isolation

Ablating the conditioning is the pre-registered K3 test. Sweeping the steering strength κ from 1 down to 0 (which is exactly the OU control) flattens the curve: the lift falls from 13.0 points (κ=1) to 0.0 points (κ=0), and the inverted-U test fails at κ=0 (Fig. 2a). Varying the barrier scale b_0 moves the optimum σ^* (Fig. 2b), as expected if the barrier geometry sets the operating point. The effect is the barrier conditioning; K3 does not fire.

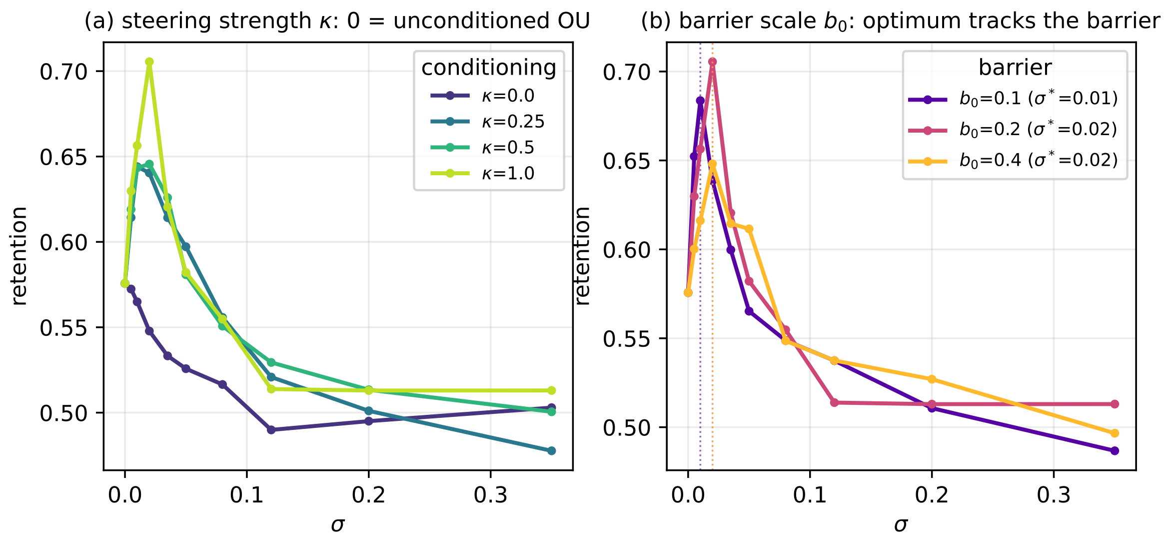

Figure 2: **Isolation.** (a) Interpolating the steering strength κ from 0 (unconditioned OU) to 1 (full h-transform): the inverted-U *emerges* with the conditioning (lift at κ=0 is 0.0 pts; at κ=1, 13.0 pts). (b) Varying the barrier scale b_0: the retention optimum σ^* tracks the barrier, evidence the optimum is set by the conditioning geometry, not a generic noise sweet-spot.

4.3 E2 — BrainScaleS-2 intrinsic-noise emulation

Re-running the sweep with the device-faithful BSS-2 noise model, the inverted-U survives (Fig. 3a): lift 11.0 points at $\sigma^* = 0.005$, colored/multiplicative/fixed-pattern noise and 6-bit weights notwithstanding. A scan over the temporal color ρ (Fig. 3b) maps the boundary. We stress the boundary of the claim: this is an *emulation*. Whether the *measured* intrinsic noise of a physical BSS-2 chip has the right structure to consolidate is kill condition K2, resolvable only on hardware, which we did not run.

4.4 E3 — matched-budget baselines

At σ^* the rule reaches retention 68.3%. Among rehearsal-free consolidation methods (which store no data) it is the best: it matches the strongest, MESU (65.2%; ours−MESU = +3.1 pts, paired Wilcoxon $p = 0.25$, not significant), and significantly beats unconditioned OU 57.4%, EWC 59.1%, Benna–Fusi 50.6% and naive 57.4% (all $p \leq 0.008$; Fig. 4a). We are explicit about the honest comparison: *plain reservoir replay*, which stores 250 raw exemplars, reaches 83.0% — 14.6 points above ours. Replay is a memory-based method in a different budget class, and, more to the point, it neither exhibits nor exploits the noise-consolidation mechanism that is our subject: our contribution is the noise→retention *signature*, and the rule's standing is realized *at its optimum*, exploiting a resource the anchored-drift methods are only harmed by. On the energy axis (Fig. 4b), generating the diffusion noise costs a digital accelerator a 23% fraction of each consolidation step; on analog silicon that fraction is physics, not compute — the dividend a von-Neumann machine must pay to fake.

4.5 E4 — second modality and the noise-optimum vs. task similarity

On continual Yin-Yang — a very different, procedural, BSS-2-native stream — the inverted-U reproduces (Fig. 5a), lift 7.4 points at $\sigma^* = 0.035$, with the OU control flat. Sweeping task similarity

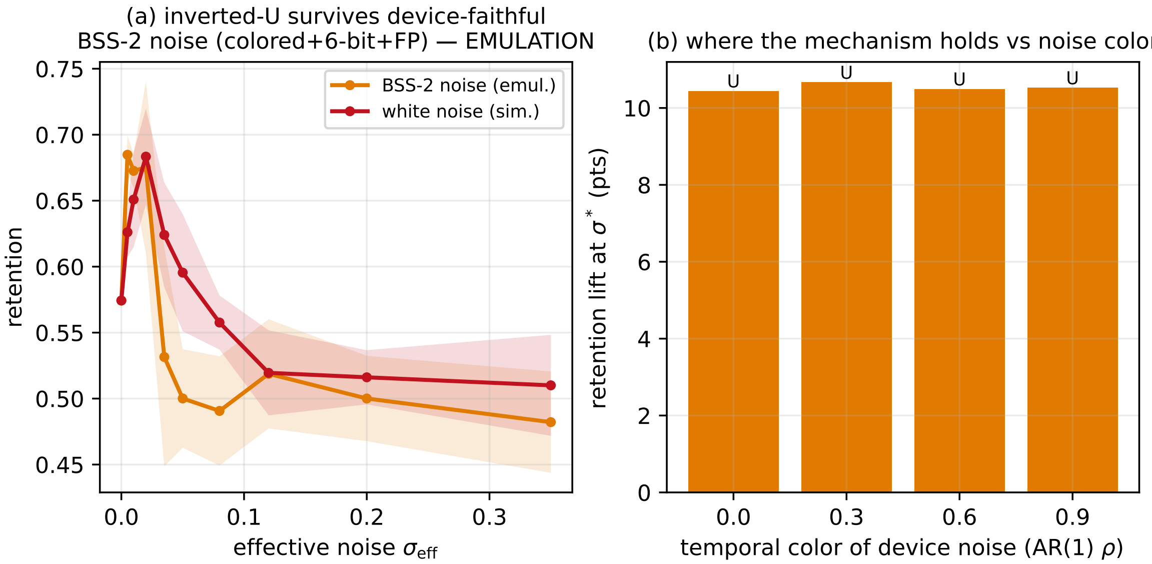

Figure 3: **Device-faithful noise (emulation, not silicon).** (a) With the diffusion replaced by the BSS-2 model (colored + multiplicative + fixed-pattern + 6-bit), the inverted-U survives (lift 11.0 pts at $\sigma^* = 0.005$). (b) Scanning the temporal color ρ of the device noise maps where the mechanism holds — the concrete content of kill condition K2, to be settled on silicon.

(Fig. 5b) shows the optimum shifting with interference, sketching where noise-as-consolidator is most useful. Figure 6 locates the mechanism: the retention optimum coincides with a *forgetting minimum* at σ^*, while plasticity (accuracy just after learning each task) is roughly noise-insensitive — the per-synapse steering suppresses interference with old memories without blocking new learning. The inverted-U is thus a protection effect: too little noise under-powers the σ^2 steering, too much lets the raw diffusion overwhelm it.

5 What we do not claim

- **The drift is not novel.** $-s(w - \mu)$ is a re-derived limit of OUA/MESU/EWC [Garcia Fernandez et al., 2024, Bonnet et al., 2025, Kirkpatrick et al., 2017]. Our contribution is the Doob barrier-conditioning (a) and the inverted-U (b), only in conjunction.
- **No measured silicon.** The BSS-2 result is a device-faithful *emulation*. We report no measured on-chip retention and no measured joules; the energy numbers are an operation-count model. On-silicon validation (K2) is pre-registered and remains.
- **Benefit only at the optimum.** We do not claim the mechanism helps at arbitrary noise; it helps in an inverted-U window and hurts outside it. We report the whole curve, including the down-slope.
- **Not a retention SOTA claim.** The result is a mechanism and a signature, not a leaderboard entry. In particular plain replay, which stores raw exemplars, retains more than our rule (83.0% vs. 68.3%); we report this openly. Our comparison of record is against *rehearsal-free* anchored-drift methods under identical noise injection, where the signature and the standing are.

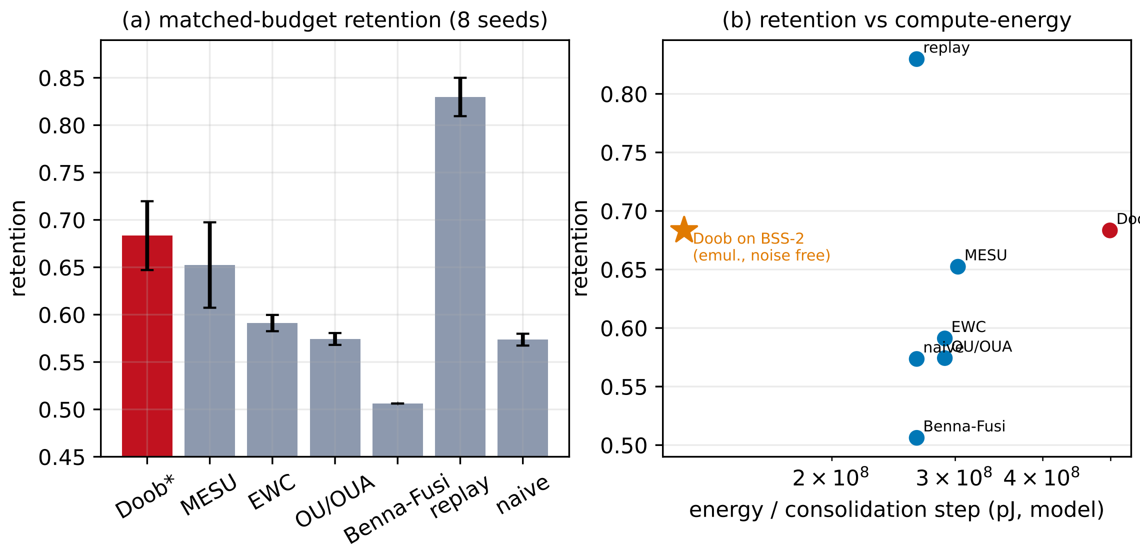

Figure 4: **Baselines (Split-MNIST, 8 seeds).** (a) At its noise optimum the barrier-conditioned rule (Doob*, 68.3%) is the strongest *rehearsal-free* method: it matches MESU (65.2%) and significantly beats OU, EWC, Benna–Fusi and naive. Plain replay stores exemplars and scores higher (83.0%) — a different, memory-based budget class, shown for reference. (b) Retention vs. modelled compute-energy: a digital accelerator pays a 23% per-step energy tax to *fabricate* the diffusion noise; on BSS-2 (emulated energy) the same noise is intrinsic and free.

6 Limitations

Small single-head MLP testbeds; a diagonal-Fisher barrier; a ground-state (infinite horizon) h-transform rather than a finite-horizon survival function; a finite-force cap that softens the exact barrier divergence; an emulated rather than measured device. Each is a place the on-silicon step (K2) or a larger study could overturn or extend the result; the pre-registered kill conditions name the failure modes.

7 Reproducibility

All code, the pre-registered `PLAN.md` (committed before results), the committed `results/*.json`, and the figure/number generators are released. Every number in this paper is a machine-generated macro (`numbers.tex` from `gen_paper_numbers.py`); `verify_regen.py` checks that `numbers.tex` is byte-identical on regeneration. Seeds, the noise grid, the operating point, and the inverted-U test are fixed in `PLAN.md`. DOI at submission.

8 Conclusion

Casting per-synapse consolidation as a Doob h-transform makes a sharp, falsifiable prediction — intrinsic noise should non-monotonically improve retention — that the anchored-drift methods whose drift we share cannot make. The prediction holds, in simulation and in a device-faithful emulation, and is isolated to the barrier conditioning. If it survives on silicon, intrinsic analog noise becomes a consolidation resource that gets *cheaper* as devices get noisier — the opposite of the usual tax. That last step is the one we have not taken, and we have been careful to say so.

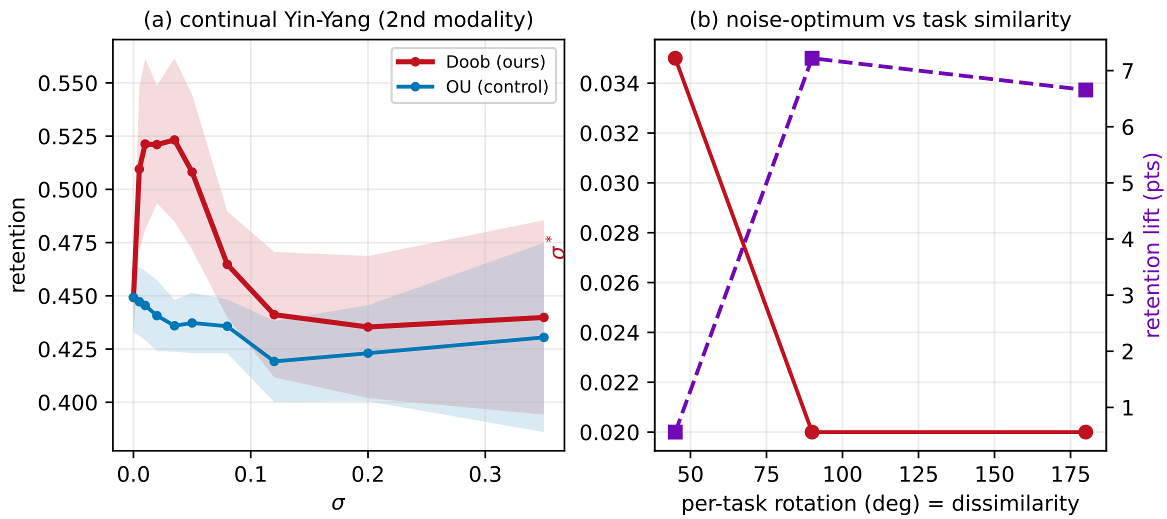

Figure 5: **Second modality.** (a) On continual Yin-Yang the inverted-U reproduces (lift 7.4 pts at $\sigma^* = 0.035$); the OU control is flat. (b) As tasks become less similar (larger per-task rotation, more interference) the noise optimum shifts, mapping the mechanism's operating regime.

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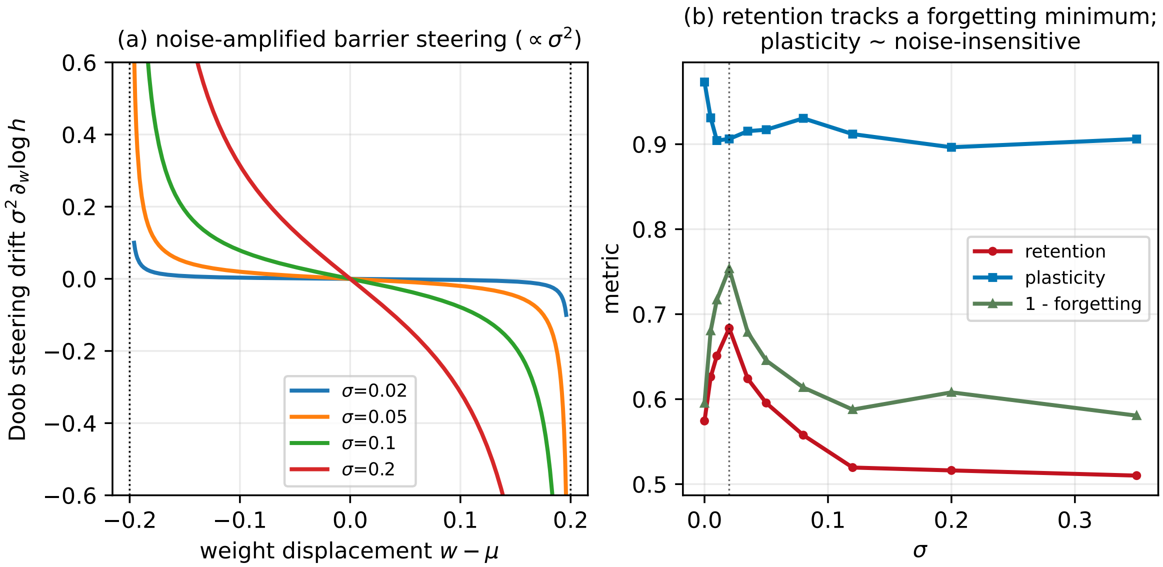

Figure 6: **Mechanism.** (a) The Doob steering drift $\sigma^2 \partial_w \log h$ for several noise levels: a barrier-divergent restoring force whose strength scales with the noise variance. (b) The retention optimum coincides with a forgetting minimum (1−forgetting peak) at σ^*; plasticity is roughly noise-insensitive. The inverted-U is a protection effect, not a plasticity tradeoff.

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